

Calculus II Lesson 06

1) Find $\int x \csc^{-1}(x^2 + 16) dx$ Hint: Let $u = x^2 + 16$; then use Formula 106.

a) $du = ?$

b) $\int x \csc^{-1}(x^2 + 16) dx =$ _____

2) Find $\int \frac{1}{x^2 \sqrt{x^2 - 25}} dx$

Hint: Use Formula 58

$$\int \frac{1}{x^2 \sqrt{x^2 - 25}} dx = \underline{\hspace{10cm}}$$

3) Find $\int \frac{14x}{(3-2x)^2} dx = 14 \int \frac{x}{9-12x+4x^2} dx$

Use Formula 29,

$$\int \frac{14x}{(3-2x)^2} dx = \underline{\hspace{10cm}}$$

4) Find $\int e^x \sin^{-1}(e^x) dx$

Hint: Let $u = e^x$; then use Formula 101

$$\int e^x \sin^{-1}(e^x) dx = \underline{\hspace{10cm}}$$

5) Find $\int \frac{1}{x^2 \sqrt{5+12x^2}} dx$ Hint: Use Formula 58

a) Using Formula 58: $u = ?$ $a = ?$ $du = ?$

b) $\int \frac{1}{x^2 \sqrt{5+12x^2}} dx =$ _____

6) Find $\int \frac{\ln x}{x(12 + 7 \ln x)} dx = \int \frac{\ln x}{(12 + 7 \ln x)} \left(\frac{1}{x} \right) dx$ Hint: Let $u = \ln x$; Use Formula 27

$$\int \frac{\ln x}{x(12 + 7 \ln x)} dx = \underline{\hspace{10cm}}$$

7) Find $\int_0^{\pi/2} \frac{\sin x}{1 + \cos^2 x} dx = \int_0^{\pi/2} \frac{1}{1 + \cos^2 x} (\sin x) dx$

Hint: Let $u = \cos x$; Use Formula 45

$$\int_0^{\pi/2} \frac{\cos x}{1 + \sin^2 x} dx = \underline{\hspace{2cm}} ?$$

8) Find the area of the region bounded by $y = \sin x + 5$, $y = 0.5x$, $x = 0$ and $x = 4$.

$$\text{Area} = \int_0^4 [(\sin x + 5) - (0.5x)] dx = \underline{\hspace{2cm}} ?$$

